

Lucky Verma

MACHINE LEARNING ENGINEER | AGENTIC SYSTEMS | KNOWLEDGE GRAPHS | RAG

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SUMMARY

Machine Learning Engineer with 4+ years building production agentic AI systems, knowledge graphs, and deep learning pipelines for large-scale data infrastructure. Public research record includes two arXiv preprints with code/artifacts. Expert in multi-agent orchestration (LangGraph, DSPy), RAG pipelines, entity resolution, model evaluation, and real-time inference serving on GPU clusters.

TECHNICAL SKILLS

AI / LLM	RAG, Agentic AI, LangGraph , DSPy , LangChain, vLLM, Fine-tuning (LoRA, RLHF/DPO), Tool Use, HF Transformers
ML / DL	PyTorch , TensorFlow, FSDP, DeepSpeed, SLURM Multi-GPU, Triton Kernels, Deep Learning , Computer Vision , NLP, CUDA
Infra	Knowledge Graphs (Apache AGE, LightRAG), PostgreSQL/pgvector, FastAPI , Docker , K8s, Terraform, AWS , Redis, Model Monitoring
Languages	Python , SQL, C++, JavaScript, Bash

WORK EXPERIENCE

Eccalon LLC (AI and federal data systems) MACHINE LEARNING ENGINEER	Nashville, TN Jun 2022 – Present
- Architected and deployed deep learning + agentic AI workflows for regulated document analysis, evidence retrieval, and corporate ownership reasoning, using LangGraph orchestration and DSPy optimization across federal and corporate data sources. - Engineered knowledge graph infrastructure (PostgreSQL + Apache AGE) with entity resolution across SAM.gov, SEC EDGAR, and GLEIF corporate registries, supporting multi-hop ownership-chain tracing with low-latency query response. - Built document intelligence pipeline with GPU-accelerated embeddings (Ollama + NIM fallback), LightRAG/PostgreSQL-backed retrieval, and source-grounded synthesis across federal databases and regulatory documents. - Built model evaluation framework for LLM baselines, feature engineering, contamination checks, and multi-annotator validation in ownership and compliance-oriented reasoning workflows. - Deployed RAG pipelines on AWS with multi-path reasoning, hybrid vector/keyword search, and grounded retrieval; built GPU-accelerated CV pipelines with PyTorch and CUDA. - Owned end-to-end ML infrastructure: Docker service orchestration, Terraform IaC, CI/CD, model serving, monitoring, and multi-backend embedding fallback for production ML services.	
University of Maryland, Baltimore County GRADUATE RESEARCH ASSISTANT	Baltimore, MD Jan 2022 – May 2023
Developed ensemble 1D U-Net for biomedical signal classification (PPG → sleep staging), achieving 92% accuracy; built web-based data science learning platform using CV/ML pipelines with pandas and Streamlit.	
Vast Dream Group AI SPECIALIST	Mumbai, India (Remote) Jan 2021 – Aug 2021
Built recommendation engine using BART zero-shot classification (96% accuracy on MNLI); architected AI measurement systems for optics industry, contributing to \$1M+ revenue pipeline across 3 product lines.	
TandM Techlabs SOFTWARE DEVELOPER	Secunderabad, India Jun 2018 – Dec 2020
Built urban mobility platform with geospatial ML: Pandana accessibility metrics, Valhalla/OSMnx route optimization on PostGIS, cached with Redis and containerized on Kubernetes, serving real-time discovery for B2B logistics.	

RESEARCH & OPEN SOURCE

When Does Removing LayerNorm Help? <i>Activation bounding Transformer normalization Reproducible artifacts</i>	arXiv Code
Solo-author public preprint studying regime-dependent behavior when replacing LayerNorm with bounded activations; released configs, result manifests, validation scripts, and artifact notes.	
When Top-1 Fails: Calibrating LoRA Monitors for Masked Diffusion LMs <i>Masked diffusion LMs LoRA monitor calibration Released aggregate artifacts</i>	arXiv Code
Co-authored public preprint with reference logging/checking scripts and sanitized aggregate result artifacts backing paper tables and figures.	
Weight Decay Regimes in Grokking Transformers <i>Grokking diagnostics Transformer training behavior Public dataset</i>	arXiv Code
Solo-author public preprint with diagnostics code, aggregate JSONs, provenance manifests, Lean 4 checks, and a Hugging Face dataset.	
Public ML and developer-tool repositories <i>97 public repos 50 non-fork public repos 126 total stars as of Jun 2026</i>	GitHub
Selected public work spans LayoutLM document classification (35 stars), driver-license OCR (26), PPG/EEG biomedical signal projects, MWB Linux, Mail Reclaim, and InfiniteContext-1B.	

EDUCATION

University of Maryland, Baltimore County <i>Master of Science in Computer Science – GPA: 3.97</i>	<i>Baltimore, MD</i> <i>May 2023</i>
• DARPA Catalyst Award (\$50K) – Caselet, selected from 250+ teams for AI-powered national security workforce tools	
SRM University <i>Bachelor of Technology in Electrical & Electronics Engineering – GPA: 3.9</i>	<i>Chennai, India</i> <i>May 2019</i>